FREEDOM INTERNATIONAL SCHOOL

WORKSHEET-MCQ (SOLUTIONS)

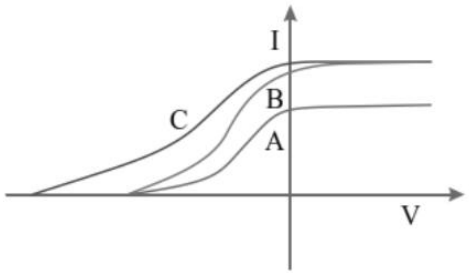
PHYSICS

CLASS XII

DUAL NATURE OF RADIATION AND MATTER

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1. In a photoelectric current experiment anode potential is plotted against plate current



(a) A and B will have same intensities while B and C will have different frequencies
(b) B and C will have different intensities while A and B will have different frequencies
(c) A and B will have different intensities while B and C will have equal frequencies
(d) B and C will have equal intensities while A and B will have same frequencies

Ans:(d) B and C will have equal intensities while A and B will have same frequencies

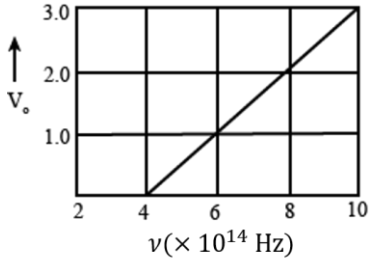
2. The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy 6 eV fall on it is 4 eV. The stopping potential, in volt, is

(a) 2 (b) 4 (c) 6 (d) 10

Ans: (b) 4

$eV_0 = 4$ eV

3. The stopping potential V_0 (in volt) as a function of frequency (υ) for a sodium emitter, is shown in the figure. The work function of sodium, from the data plotted in the figure, will be
(Given, h= 6.63 x 10^{-34} Js, e= 1.6 x 10^{-19} C)



(a) 1.95 eV (b) 2.12 eV (c) 1.82 eV (d) 1.66 eV

Ans: (d) 1.66 eV

Threshold frequency, $v_0 = 4 \times 10^{14}$ Hz

$\Rightarrow$ Work function $\phi = hv_0$

$\Rightarrow \phi = 6.63 \times 10^{-34} \times (4 \times 10^{14})$ J

$$\Rightarrow \phi = \frac{(6.63 \times 10^{-34}) \times (4 \times 10^{14})}{1.6 \times 10^{-19}} \text{eV}$$

$\Rightarrow \phi = 1.66$ eV

4. If the kinetic energy of free electrons doubles, its de- Broglie wavelength changes by the factor
   (a) 1/2 (b) 2 (c) $1/\sqrt{2}$ (d) $\sqrt{2}$
   Ans: (c) $1/\sqrt{2}$

$$\lambda = \frac{h}{\sqrt{2m(KE)}}$$

Or $\lambda \propto \frac{1}{\sqrt{(KE)}}$.

So if we double the kinetic energy,

$$\lambda \propto \frac{1}{\sqrt{2(KE)}}$$

Or $\lambda \propto \frac{1}{\sqrt{2}}$

5. An electron of mass m and charge e is accelerated from rest through a potential difference V in vacuum. Its final velocity will be
   (a) $\sqrt{\frac{2eV}{m}}$ (b) $\sqrt{\frac{eV}{m}}$ (c) $\frac{eV}{2m}$ (d) $\frac{eV}{m}$

   Ans: (a) $\sqrt{\frac{2eV}{m}}$

   ½ $mv^2 = eV$

6. If a photon has velocity c and frequency υ, then which of the following represents its wavelength?
   (a) $\frac{h\nu}{c^2}$ (b) hυ (c) $\frac{hc}{E}$ (d) $\frac{h\nu}{c}$

   Ans: (c) $\frac{hc}{E}$

7. When ultraviolet rays are incident on a metal plate then photoelectric effect does not occur, it occurs by the incidence of
   (a) infrared rays (b) X- rays
   (c) radio wave (d) micro wave

Ans: (b) X- rays

8. When light of wavelength 300 nm falls on a photoelectric emitter, photoelectrons are liberated. For another emitter, however, light of 600 nm wavelength is sufficient for creating photoemission. What is the ratio of the work functions of the two emitters?

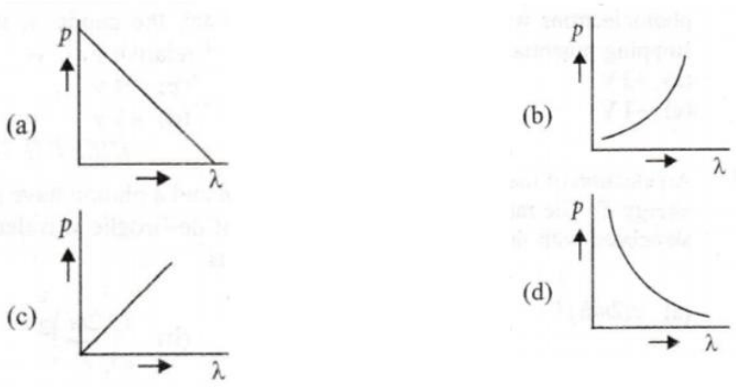
(a) 1 : 2 (b) 2 : 1 (c) 4 : 1 (d) 1 : 4

Ans: (b) 2 : 1

Work function = $\frac{hc}{\lambda_0}$ where λ_0 is threshold wavelength.

$\therefore \frac{W_{O_1}}{W_{O_2}} = \frac{\lambda_{O_2}}{\lambda_{O_1}} = \frac{2}{1}$

9. Which of the following figures represent the variation of particle momentum associated with de- Broglie wavelength?



Ans: (d)

10. When photons of energy, hυ fall on an aluminium plate (of work function E_0), photoelectrons of maximum kinetic energy K are ejected. If the frequency of radiation is doubled, the maximum kinetic energy of the ejected photoelectrons will be

(a) K+ hυ (b) K+ E_0 (c) 2 K (d) K

Ans: (a) K+ hυ

$K = h\nu - E_0$

$h\nu = K + E_0$ ---(1)

- When the frequency is doubled the above equation becomes

$2h\nu = KE_2 + E_0$

Substituting the value of E_0 from equation 1

$2h\nu = KE_2 + (h\nu - K)$

$2h\nu - h\nu + K = KE_2$

$h\nu + K = KE_2$

$KE_2 = h\nu + K$

For questions 11 to 15, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

11. **Assertion:** In the process of photoelectric emission, all emitted electrons do not have the same kinetic energy.
Reason: If radiation falling on the photosensitive surface of a metal consists of different wavelengths, then the energy acquired by electrons absorbing photons of different wavelengths shall be different.
Ans: B
12. **Assertion:** Photoelectric saturation current increases with the increase in frequency of incident light.
Reason: The energy of incident photons increases with increase in frequency and as a result photoelectric current increases.
Ans: D
13. **Assertion:** Photoelectric effect demonstrates the wave nature of light.
Reason: The number of photoelectrons is proportional to the frequency of light.
Ans: D
14. **Assertion:** The energy (E) and momentum (p) of a photon are related by p= E/c.
Reason: The photon behaves like a particle.
Ans: **A**
15. **Assertion:** The mass of a moving photon varies inversely as the wavelength.
Reason: Energy of the particle = mass x (speed of light)^2
Ans: B